

COMPUTER NETWORK TECHNOLOGY (300)

NATIONAL – 2019

Knowledge Test:

Certification Exam: MTA Networking Fundamentals _____ (100 points)

Objective Test:

Objective Test: _____ (500 points)

TOTAL POINTS _____ ***(600 points)***

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

No more than sixty (60) minutes testing time

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Identify the letter of the choice that best completes the statement or answers the question. Mark A if the statement is True. Mark B if the statement is False.

1. At what layer of the OSI model does a firewall operate?
 - a. Layer 1
 - b. Layer 2
 - c. Layer 3
 - d. Layer 7
2. A honey pot would be an example of what type of network device?
 - a. Firewall
 - b. IDS/IPS
 - c. Content filter
 - d. Load balancer
3. What network service hides the internal addressing in use from external clients?
 - a. DHCP
 - b. DNS
 - c. RAS
 - d. NAT
4. What device acts as an intermediary for requests from clients seeking resources from other servers?
 - a. DNS
 - b. Proxy server
 - c. Content filter
 - d. Load balancer
5. Which of the following is an example of a circuit-switched technology?
 - a. ATM
 - b. Frame Relay
 - c. Wireless
 - d. ISDN
6. Which of the following is considered Layer 2.5 technology?
 - a. MPLS
 - b. DSL
 - c. Satellite
 - d. Broadband cable
7. An excellent option for WAN connectivity in a very remote area of the globe is _____.
 - a. DSL
 - b. Satellite
 - c. Cable Broadband
 - d. Metro-Ethernet

8. What is an example of a packet-switched technology?
 - a. T1
 - b. ATM
 - c. Frame Relay
 - d. ISDN
9. What tool can be used to determine exactly where a break exists in a cable?
 - a. Crimper
 - b. Snip
 - c. Wire stripper
 - d. TDR
10. What category of UTP supports twice as many frequencies as its previous version and can be used for 10GBASE-T networks, which can transmit data at a rate of 10 billion bits per second (10 Gbps)?
 - a. Cat 5
 - b. Cat 6
 - c. Cat 6a
 - d. Cat 5e
11. The type of network featured in very early Ethernet implementations was which of the following?
 - a. Ring
 - b. Bus
 - c. Star
 - d. Dual ring
12. A point-to-multipoint network is often called?
 - a. Client-server
 - b. Full mesh
 - c. Hub-and-spoke
 - d. Peer-to-peer
13. The class of IP address 172.16.24.0/24 is _____.
 - a. Class A
 - b. Class B
 - c. Class C
 - d. Class D
14. Which of the following is *not* a valid traffic form in IPv6?
 - a. Unicast
 - b. Link local
 - c. Anycast
 - d. Broadcast

15. What routing protocol characteristic indicates the believability of the routing protocol (as opposed to other routing protocols)?
- Weight
 - Metric
 - Administrative distance
 - SPF algorithm
16. What is the admin distance for an unreachable route?
- 90
 - 120
 - 255
 - 170
17. What is the admin distance for RIP?
- 90
 - 120
 - 110
 - 170
18. What set of tools in QoS seeks to smooth out traffic flows?
- Traffic shaping
 - Link efficiency
 - Congestion management
 - Congestion avoidance
19. What component in a VoIP network is now considered legacy?
- Call agent
 - PBX
 - IP phone
 - SIP agent
20. What protocol in a VoIP network is used to carry voice packets?
- Call agent
 - Gateway
 - IP phone
 - RTP
21. Which of the following is the *most likely* reason a company would install a wired network instead of a wireless one?
- Performance
 - Manageability
 - Cost
 - Administrative overhead

22. What type of SNMP communication is unsolicited from a managed device?
- GET
 - GETNEXT
 - TRAP
 - SET
23. If there is a key WAN link in your company that many systems must traverse at one point or another in the day, this network area might become a _____.
- baseline
 - bottleneck
 - reverse route
 - trend
24. Your company is considering changing several key technologies from the application to the network layer. What would you recommend?
- Security auditing
 - Performance metric testing
 - Compliance testing
 - A testing lab
25. What performance optimization technique is often typical with VLANs?
- Increasing the number of host systems
 - Decreasing the number of host systems
 - Bunching top talkers
 - Implementing port security
26. Which updates most often *require* a device reboot?
- App Updates
 - Firmware
 - Minor updates
 - Patches
27. Why might you assign a default gateway address to a Layer 2 switch?
- To permit the switch to communicate in the local Layer 2 domain
 - To permit the switch to route
 - To permit the switch to reach remote admins
 - To permit the switch to trunk
28. You have Port Security configured on your switch port to lock it to a single MAC address that you have statically configured. You unplug your computer and plug in an IP phone. What happens by default?
- The switch port blocks the phone but would permit the computer
 - The switch port is shut down
 - The switch port permits the traffic, but alerts you
 - The switch port redirects the traffic

29. Refer to the exhibit in the box below. What is the problem with the access list shown?

```
access-list 10 permit 10.0.0.0 0.255.255.255
access-list 10 deny host 10.1.1.1
access-list 10 permit any
```

- a. The access list is permitting the host traffic before denying it
 - b. The access list cannot have more than two entries
 - c. The access list must end with a deny statement
 - d. The access list is fine
30. What type of procedure ensures that you can restore the network to the way it was before you implemented some troubleshooting-based configurations?
- a. Disaster recovery
 - b. System restore
 - c. Rollback
 - d. Last Known Good
31. What device would you use to detect opens and shorts?
- a. Toner probe
 - b. Cable tester
 - c. Wi-Fi analyzer
 - d. Speed test site
32. Refer to the exhibit in the box below. What command produced the following results?

```
Active Internet connections
Proto Recv-Q Send-Q Local Address           Foreign Address
(state)
tcp4          0      0 192.168.0.3.59571      server-54-192-
16.https ESTABLISHED
tcp4          0      0 192.168.0.3.59570      ec2-23-21-245-
```

- a. nbtstat
 - b. traceroute
 - c. pathping
 - d. netstat
33. If an attacker is using a rogue access point to try to launch attacks in your network, what will they most likely do?
- a. Have the access point in a unique SSID
 - b. Have the access point in your SSID
 - c. Use WPA2 on the AP
 - d. Feature the AP in an overlapping channel

34. Before placing the APs in a large corporate office, you should perform which of the following?
- Cable test
 - Security audit
 - Site survey
 - Physical check
35. Using a cable tester to discover that pin 1 connects to pin 2 in your cable is called which of the following?
- Open
 - Attenuation
 - Crosstalk
 - Short
36. What is a rogue access point?
- An access point that is not authorized by your organization
 - An access point that is misconfigured
 - An access point for another SSID
 - An access point for another channel
37. Which of the following should you protect against when adding a new AP to extend the reach of your SSID?
- Reconnaissance system
 - Man-in-the-middle system
 - Botnets
 - Worms
38. What is a typical result of an IP address conflict on the network?
- The inability to communicate locally without impacting remote communications
 - An error indication in the OS
 - A failure of only one system to communicate
 - No impact whatsoever
39. Logging in to the network only once is known as what?
- SSH
 - SSL
 - SSO
 - SSH
40. The MAC address is associated with which layer of the OSI model?
- Layer 1
 - Layer 2
 - Layer 3
 - Layer 4

41. What transmission method does 802.11b use?
 - a. MU-MIMO
 - b. MIMO
 - c. OFDM
 - d. DSSS
42. What numbering system do we use in our everyday life?
 - a. Base 2
 - b. Base 8
 - c. Base 10
 - d. Base 16
43. The ping utility relies on what protocols?
 - a. ICMP
 - b. IGMP
 - c. PIM
 - d. TCP
44. What frequency band does 802.11a use?
 - a. 2.4 GHz
 - b. 5 GHz
 - c. 10 GHz
 - d. 4 GHz
45. What is the distance limitation for 10BASE-2?
 - a. 40 km
 - b. 100 m
 - c. 185 m
 - d. 300 m
46. What is the approximate max speed of 802.11n?
 - a. 11 Mbps
 - b. 54 Mbps
 - c. 100 Mbps
 - d. 300 Mbps
47. What servicing of a data center environment might involve chemicals, gas, and electricity?
 - a. Fire suppression
 - b. HVAC
 - c. Power
 - d. Load

48. What frequency band does 802.11g use?
- a. 4 GHz
 - b. 5 GHz
 - c. 2.4 GHz
 - d. 10 GHz
49. What protocol should be used to remote to a Windows desktop from another location?
- a. SIP
 - b. RTP
 - c. RDP
 - d. SNMP
50. Which is a DNS record for an e-mail server?
- a. CNAME
 - b. SOA
 - c. NS
 - d. MX